

SC3103: Embedded Programming

Tutorial 2: Real-Time Schedulability

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1. Consider the following periodic real-time task set:

$$\tau_1 = (3, 1, 3), \tau_2 = (5, 1, 5), \tau_3 = (7, 3, 7).$$

Which of the following scheduling algorithms, EDF or RM, can successfully schedule this task set. Justify your answer.

2. Consider the following sporadic real-time task set:

$$\tau_1 = (15, 5, 15), \tau_2 = (17, 5, 17), \tau_3 = (19, 5, 19).$$

Is it schedulable under RM? Justify your answer

3. Consider a task system comprising four tasks, $\tau_1 = (4, 1, 4)$, $\tau_2 = (5, 3, 5)$, $\tau_3 = (6, 2, 6)$ and $\tau_4 = (8, 5, 8)$, and scheduled on a **dual-core** platform.

- (a) Is this task system feasible? Justify your answer.
- (b) Identify all possible partitions of this set of tasks such that the resulting system is schedulable under both Partitioned Earliest Deadline First (EDF) and Partitioned Rate Monotonic (RM). Justify your answer.
- (c) Is this task system schedulable under Global Rate Monotonic (RM) scheduling algorithm? Justify your answer.
- (d) Construct the Global Earliest Deadline First (EDF) schedule for this task system up to time instant 15. Can you conclude from this schedule whether the task system is schedulable under Global EDF? Justify your answer.